ECON 7201
Applied Econometrics
Assignment 2

Due Date

November 5, 2026 at 11:59 PM

Directions

Answer all questions. Upload both a PDF and Quarto file to your website in a new blog post.

Git and GitHub

1. (a) Create a new R project in your **econ_3201** directory called **assignment_2**.
   (b) Download the assignment PDF and Quarto file the **assignment_2** folder.
   (c) Commit and push the changes to your **econ_3201** repository on GitHub.com.

LaTeX

Matrices are created in LaTeX using the `\begin{bmatrix}...\end{bmatrix}` command. To separate entries along the same row, use `&`. To end a line, use `\\`. To make vertical elipses (:), use `\vdots`. Practice writing the following matrices and vectors in LaTeX. Write the following matrices in LaTeX.

2. (a)

> Tip
>
> $$X'X = \begin{bmatrix} n & \sum_{i=1}^{n} x1i & \sum_{i=1}^{n} x_{2i} \\ \sum_{i=1}^{n} x_{1i} & \sum_{i=1}^{n} x_{1i}^2 & \sum_{i=1}^{n} x_{1i}x_{2i} \\ \sum_{i=1}^{n} x_{2i} & \sum_{i=1}^{n} x_{1i}x_{2i} & \sum_{i=1}^{n} x_{2i}^2 \end{bmatrix}$$

(b)

Tip

$$\Omega = \begin{bmatrix} \sigma_1^2 & 0 & 0 & 0 \\ 0 & \sigma_2^2 & 0 & 0 \\ 0 & 0 & \sigma_3^2 & 0 \\ 0 & 0 & 0 & \sigma_4^2 \end{bmatrix}$$

R

3. In this question we compare standard errors based on (incorrect) asymptotic assumptions with those based on alternate (appropriate) estimator (White). Consider one sample drawn from the following data generating process (DGP) which we will simulate in R:

```
set.seed(123)
n <- 25
x <- rnorm(n,mean=0.0,sd=1.0)
beta0 <- 1
beta1 <- 0
## x is irrelevant in this model, the data generating process is as follows:
dgp <- beta0 + beta1*x
## The residual is heteroskedastic by construction
e <- x^2*rnorm(n,mean=0.0,sd=1.0)
y <- dgp + e
```

(a) Compute the OLS estimator of β_2 and its standard error using the `lm()` command in R for the model $y_i = \beta_1 + \beta_2 x_i + \epsilon_i$ based on the DGP given above.

Tip

```
ols_model <- lm(y ~ x)

summary (ols_model)
```

The OLS estimate of beta_2 is 0.4610, and it's standard error is 0.2563.

(b) Next, compute the standard error of $\hat{\beta}_2$ by computing $\hat{\sigma}^2(X'X)^{-1}$ in R using matrix commands, and verify that the two standard error estimates are identical.

Tip

```
# Create the X matrix
X <- cbind(1, x)

# Calculate the OLS residuals
beta_hat <- solve(t(X) %*% X) %*% t(X) %*% y
e_hat <- y - X %*% beta_hat

# Calculate estimated error variance
sigma2_hat <- as.numeric(t(e_hat) %*% e_hat) / (n - 2)

# Calculate variance-covariance matrix
V  <- sigma2_hat * solve(t(X) %*% X)

# Calculate standard error of beta2
sqrt(V[2, 2])
```

The standard error of beta_2 calculated using the matrix formula is approximately 0.2563. This is identical to the standard error obtained using the lm() function in question 3(a).

(c) Compute White's heteroskedasticity consistent covariance matrix estimator using matrices in R and report the White estimator of the standard error of $\hat{\beta}_2$. Compare this with that from 3 (a) above.

Tip

```
# Calculate White heteroskedasticity-consistent covariance matrix

bread <- solve(t(X) %*% X)

meat <- t(X) %*% (X * as.vector(e_hat^2))

V_white <- bread %*% meat %*% bread

# Calculate White standard error of beta2

se_white <- sqrt(V_white[2, 2])

se_white
```

The white heteroskedasticity-consistent standard error of beta_2 is 0.4529. The standard

errors in 3(a) are identical. The White standard error is larger, reflecting the adjustment for heteroskedasticity.

4. Let $\hat{\theta}$ be an estimator for the population parameter θ. $\hat{\theta}$ is said to be unbiased if $E(\hat{\theta}) = \theta$. That is, if the mean of the sampling distribution of $\hat{\theta}$ is equal to the true population value.

   Consider the model

   $$y_i = \beta_0 + \beta_1 x_{1,i} - \beta_2 x_{2,i} + \epsilon_i.$$

   Lets provide empirical evidence that the ordinary least squares estimators $\hat{\beta}_0$, $\hat{\beta}_1$, and $\hat{\beta}_2$ are unbiased estimators of β_0, β_1, β_2, respectively, using R.

   (a) Set the seed to 1, i.e., `set.seed(1)`.
   (b) Set the number of observations $n = 100$
   (c) Generate the following model

   $$y_i = 2 + 3.5x_{1,i} - 9.2x_{2,i} + \epsilon_i,$$

   where $x_1 \sim N(3, 6)$, $x_2 \sim N(2, 4)$, and $\epsilon \sim N(0, 100)$. To create a normally distributed variable, use the `rnorm(n, mean, sd)` command in R.
   (d) Estimate the model coefficients using the `lm()` command. (Search `?lm()` in the console for more info).
   (e) Using a `for()` loop, replicate the model above $M = 1000$ times and save the coefficient estimates from each iteration.
   (f) Using `hist()`, plot the sampling distributions of the coefficient estimates, β_1 and β_2.
   (g) Add a vertical line to each figure at the mean of the respective variable. Search `?abline()` in your console.

Tip

```r
# 4(a) Set the seed
set.seed(1)

# 4(b) Number of observations
n <- 100

# 4(c) Generate the explanatory variables and error term
x1 <- rnorm(n, mean = 3, sd = 6)
x2 <- rnorm(n, mean = 2, sd = 4)
epsilon <- rnorm(n, mean = 0, sd =100)

# Generate the dependent variable
y <- 2 + 3.5*1 - 9.2*x2 + epsilon

# 4(d) Estimate the model using OLS
model <- lm(y ~ x1 + x2)

summary(model)
```

```r
set.seed(1)
n <- 100

# 4(e) Repeat the simulation 1000 times

M <- 1000

# Store the coefficient estimates
estimates <- matrix(NA, nrow = M, ncol = 3)

for (i in 1:M) {
  x1 <- rnorm(n, mean =3, sd = 6)
  x2 <- rnorm(n, mean =2, sd = 4)
  epsilon <- rnorm(n, mean = 0, sd = 100)

  y<- 2 + 3.5*x1 - 9.2*x2 + epsilon

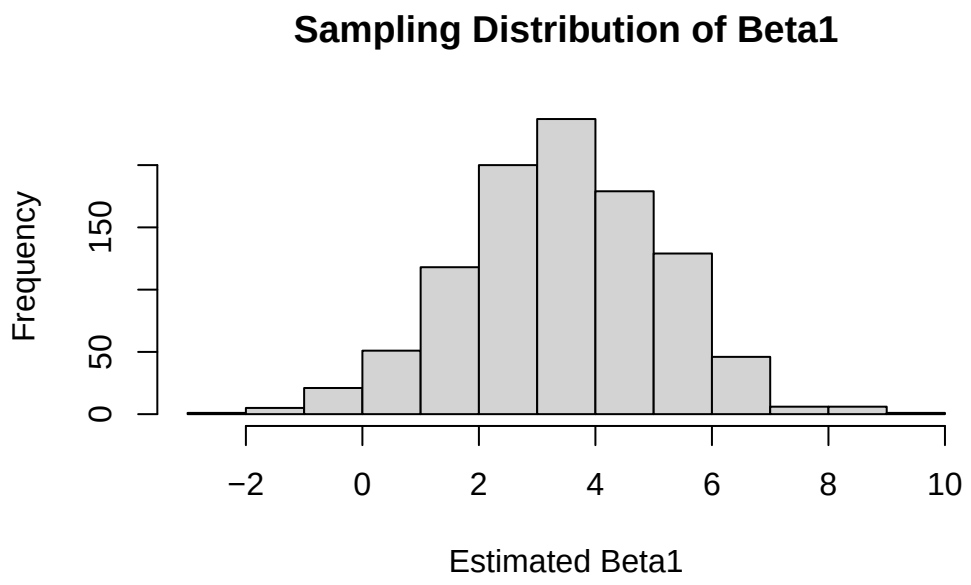
  model <- lm(y ~ x1 + x2)

  estimates[i, ] <- coef(model)
}

# Calculate the average estimates
colMeans(estimates)
```

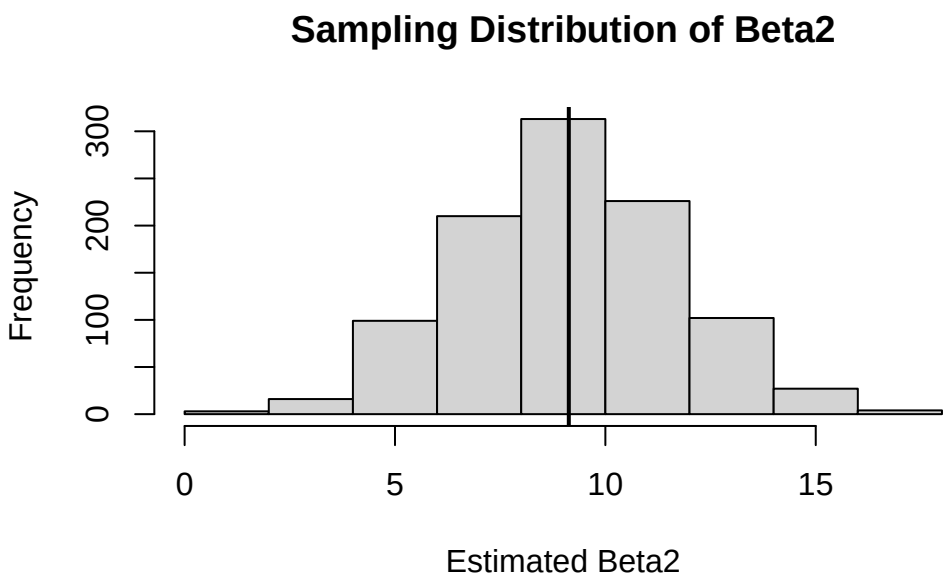
```
[1]  1.835807  3.412558 -9.131447
```

```r
# 4(f) Histogram for beta1
hist(estimates[, 2],
     main = "Sampling Distribution of Beta1",
     xlab = "Estimated Beta1")
```




```
# Histogram for beta2
hist(-estimates[, 3],
     main = "Sampling Distribution of Beta2",
      xlab = "Estimated Beta2")

abline(v = mean(-estimates[, 3]), lwd =2)
```


Sampling Distribution of Beta2
Frequency
0
100
200
300
0
5
10
15
Estimated Beta2